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Summary

I am currently a Ramón y Cajal Fellow at the department of Atomic, Molecular and Nuclear Physics of the University of Sevilla (Spain). Previously, I was a Senior Fellow at the Theory group of CERN for three years (2022-2025) and I worked in the physics department of the University of Milano (Italy) as *Assegnista di Ricerca* (4 years) in the ERC project N3PDF (P.I. Prof. S. Forte). I did my Ph.D. at Durham University (UK) under the supervision of Prof. N. Glover, where I was part of the Initial Training Network *Higgstools* and worked on the second order QCD corrections to the Higgs boson production processes (H+J and VBF-H).

My main line of research is the application of machine learning (ML) techniques and cutting-edge technologies to theoretical particle physics. Some examples of my work include the development of a GPU-cabable event generator as well as pioneering work in applying quantum ML algorithms to address important challenges in particle physics, such as multi-dimensional parametric integration or parton density estimation. Additionally, as the Research & Development coordinator of the NNPfD collaboration, I lead efforts to model the internal structure of the proton utilizing artificial intelligence.

Beyond particle physics, I am very interested on software development, system administration and open source software. Most of my work require building and testing new ML models, analyzing large amounts of data as well as interpreting the final results. During my PhD I had the opportunity to do an internship on the Projects and Technology branch of Shell

I have participated in many conferences across the world and organized three editions of the YTF (Young Theorists Forum) held in Durham around Christmas.

Research Career

University of Sevilla

Ramón y Cajal Fellow

Competitive research grant awarded by the Spanish Ministry of Science and Universities.

Sevilla (Spain)

2025-Currently

CERN

CERN Senior Fellow in the CERN theory (TH) group

Competitive research position in the QCD subgroup. Since 2024, Research & Development coordinator of NNPfD. Organizer of the weekly QCD seminars.

Geneva (Switzerland)

2022-2025

University of Milan

Assegnista di ricerca

Part of the N3PDF research project. PI Stefano Forte. Financed by the European Research Council through an Advanced Grant (n 740006) within the Horizon 2020 Research & Innovation Programme

Milan (Italy)

2018-2022

Durham University

PhD Thesis, Supervisor: Nigel Glover

Next-to-Next-to-Leading Order QCD Corrections to Higgs Boson Production in Association with two Jets in Vector Boson Fusion. Financed by the Research Executive Agency (REA) of the European Union under the Grant Agreement PITN-GA-2012-316704 ("HiggsTools")

Durham (UK)

2014-2018

University of Zurich*Academic Secondment, supervisor: Thomas Gehrmann***Zurich (Switzerland)***Oct-Dec 2016***University of Valencia & IFIC***Master in Advanced Physics: Theoretical Physics, 94.6%**Master Thesis supervisor: German Rodrigo**Study of charge asymmetry in $t\bar{t}$ production through axigluons***Valencia (Spain)***2013-2014***IFIC (Valencia)***Research Stay, master in advanced physics, Supervisor: M. Vos**Project Title: Experimental Limitations to Charge Asymmetry measurement in top quark pair production at hadron colliders***Valencia (Spain)***2014***National Accelerators Center (CNA Sevilla)***Research Stay, degree in physics, Supervisor: J.M. Lopez-Gutierrez**Project Title: Development of computing tools for the analysis of Accelerator Mass Spectrometry results at the National Accelerators Center***Seville (Spain)***June 2013***University of Seville***Degree in Physics, 82.3%**Bachelor's Thesis supervisor: Antonio Moro**Application of numerical resolution of a system with coupled differential equations to Quantum Scattering Problems with Internal Degrees of Freedom***Seville (Spain)***2009-2013*

PhD Thesis

Title: Next-to-Next-to-Leading Order QCD Corrections to Higgs Boson Production in Association with two Jets in Vector Boson Fusion**Supervisors:** Nigel Glover (Durham U.) & Thomas Gehrmann (Zurich U.)**Abstract:** In this thesis the second-order QCD corrections to electroweak production of a Higgs boson in association with two jets through vector boson fusion are considered. This calculation is fully differential in the kinematics of the Higgs boson and of the final state jets. Infrared divergences are regulated using the antenna subtraction method. We detail the implementation of the process in the parton-level Monte Carlo integrator NNLOJET and present inclusive calculations as well as differential distributions for a wide range of observables at different center-of-mass energies.**Grant:** European Union, PITN-GA-2012-316704. Higgstools Initial Training Network**URL:** <http://etheses.dur.ac.uk/12806/>

Teaching Experience

Undergraduate Course*Física Cuántica, Theory and laboratory***Universidad de Sevilla***2025-***Master Course***Quantum Mechanics, Theory and problems***Universidad de Sevilla***2025-***Advanced Artificial Intelligence for Precision High Energy Physics***Main tutor for the Machine Learning practical classes***6h***July 2023***Tutorial for the CMS collaboration***Tutorial of NNPDF fitting code***4h***February 2023***Teaching Assistant***Corso di Informatica, 3 years, 108h total***University of Milan (Italy)***2019-2022*

Teaching Assistant <i>Fisica Quantistica II, 26h</i>	University of Milan (Italy) 2020-2021
Teaching Assistant <i>Fisica Quantistica I, 10h</i>	University of Milan (Italy) 2019-2020
NNPDF Code Meeting <i>Course on the usage of the Keras and Tensorflow libraries, 5h</i>	Cambridge (UK) June 2019
Teaching Assistant <i>First Year experimental methods course, weekly exercises, 36 h</i>	Durham University (UK) 2017-2018

Complementary Education

CERN Visits Service training <i>Training on outreach and scientific communication for CERN guides</i>	CERN 2024
Cisco Networking Academy <i>Cisco Cybersecurity Scholarship</i>	Remote January-June 2021
Cisco Networking Academy <i>Introduction to Cybersecurity</i>	Remote April 2020
Xilinx Developer Forum <i>FPGA Developers Forum</i>	The Hague (The Netherlands) November 2019
ExotHiggs <i>Summer School on Higgs and BSM Physics</i>	Zuoz (Switzerland) August 2016
YETI <i>Winter School: Prospects and Challenges for LHC Run II</i>	Durham (UK) January 2016
Higgstools Summer School <i>Summer School on Higgs Physics for Early Stage Researchers</i>	Aosta Valley (Italy) July 2015
Higgstools First Young Researches Meeting <i>Teamwork, Communication and Media training</i>	Durham (UK) February 2015

Non-academic work experience

Shell (Projects & Technology Division) <i>Fortran and C Developer</i> Dutch division of the Seismic Applications team (managed by Rob Eppenga). As part of the Higgstools ITN I was given the opportunity of working at Shell for several months. In Shell I worked on the SIPMAP package, a suite of programs used for oil exploration and seismic tomography. While the formal detail of the algorithms used fall under a completely different branch of physics, the computing side was actually quite close to what it is done in high energy physics research. My task during this internship consisted on the development and maintenance of the program (the oldest pieces written in Fortran, some of the more modern features C and C++). Runs of this code are very costly and thus optimisation is key, my focus during those months was on improving some of the algorithms and streamlining the workflow of the software. I also worked on porting parts of the code to new hardware (32 bits to 64 bits and GPU accelerators).	Rijswijk (The Netherlands) 2016
FamilyApp <i>Frontend and Backend Developer, Python, HTML</i> Sole developer of both the web interface and administration backend of the service.	Seville (Spain) 2014

Academic Open-Source Software

Creating new software is a cornerstone of my work. Moreover, I firmly believe that, especially in science, all tools should be open to the wider community whenever possible. To achieve this, I have spent some considerable effort not only to make the code accessible but also to ensure it is well-documented and user-friendly. The following is a selection of open-source academic tools in which I have played a leading role:

NNPDF framework

Machine Learning, AI, Data Analysis and Visualization, PDF fitting, High Energy Physics

Released: 2021, github.com/NNPDF/nnpdf

Eur.Phys.J.C 81 (2021) 10, 958

Framework for fitting Parton Distribution Functions (PDF) using experimental data and theoretical inputs. All data and theory predictions used during the fits are also public for reproducibility. It also includes a complete data analysis and visualization suite. I am currently the main maintainer and code coordinator of the NNPDF collaboration.

MadFlow

Monte Carlo simulations, GPU computing

Released: 2021, github.com/N3PDF/madflow

doi:10.5281/zenodo.4954375

Framework for Monte Carlo simulation of particle physics processes designed to take full advantage of hardware accelerators. Processes can be generated using MadGraph5_aMCNLO and are then output in vectorized (or tensorized) form by the madflow-provided plugin.

PDFFlow

Proton physics, GPU computing

Released: 2020, github.com/N3PDF/pdfflow

doi:10.1016/j.cpc.2021.107995

Parton distribution function interpolation library written in Python and based on the TensorFlow framework. It is developed with a focus on speed and efficiency, enabling researchers to perform very expensive calculation as quick and easy as possible.

VegasFlow

Numerical calculations, GPU computing

Released: 2020, github.com/N3PDF/vegasflow

doi:10.1016/j.cpc.2020.107376

Monte Carlo integration library written in Python and based on the TensorFlow framework. It is developed with a focus on speed and efficiency, enabling researchers to perform very expensive calculation as quick and easy as possible.

pyHepGrid

Python, grid computing

Released: 2016, github.com/scarlehoff/pyHepGrid

doi:10.5281/zenodo.3233861

Core developer of the pyHepGrid tool for distributed computing. Used to run in a systematic and coherent manner resource-hungry programs typically used for HEP simulations. The development of pyHepGrid was done with the focus on NNLOJET but has since being extended successfully to also run other programs such as MCFM, Sherpa or HEJ.

Relevant computer skills

Programming Languages: Fortran, Python, C, **Operating System:** Linux, MacOS, Windows
C++, js, OpenCL, Cuda

Scripting/Macro Languages: Bash, Latex, gnu-plot **Computing Tools:** Maple, Mathematica, Matlab, Grid Computing

ML Libraries: Keras, Tensorflow, PyTorch, pandas, scipy **Technologies:** Quantum Computing, multiprocessing, GPU programming

Quantum computing Libraries: Qibo, Qiskit **HEP Tools:** Madgraph, root, Pythia

Other Projects

pybliotecario

github.com/scarlehoff/pybliotecario

Bot in python that use different remote APIs such as Facebook Messenger API or Telegram to open a communication channel between the social messaging system of choice and the server.

Python, messaging bot

Currently

Open Source

github.com/scarlehoff

I often contribute in different open source projects and am currently the maintainer of several packages in conda-forge and the Arch User Repository

Open source contributor

Currently

Management Experience

QCDLHC 2025

Conveener

Stony Brook (USA)

2025

Management & Mentoring

Code Coordinator of the NNPDF collaboration

2024-currently

AIEP School

Member of the organising committee

Lake Como School

2023,2025

Seminar organizer

Organizer of the weekly QCD seminar

CERN

2023-2025

Seminar organizer

Organizer of the joint Milan-Bicocca & Milan weekly seminars

Milan (Italy)

2022

HiggsTools Final Meeting

Member of the organising Committee

Durham (UK)

September 2017

Annual YTF (Young Theorist Forum) 8, 9, 10

Member of the organising Committee

Durham (UK)

2016-2018

ICHEP 2014

Outreach activities

Valencia (Spain)

July 2014

Participation in grants

2025-2029: Ramón y Cajal

Tenure Track position by the Spanish Ministry of ScienceInnovation and Universities (approx 250k€)

Automate Monte Carlo simulation on hardware accelerators **University of Milan (Italy)**

Linea 2A, 15000€ (4 Co-Authors)

2020-2021

New hardware for HEP

Linea 2A, 6000€ (3 Co-Authors)

University of Milan (Italy)

2019-2020

Awards & Accreditations

2025: Profesor Contratado Doctor: "Associate Professor" level recognized by the Spanish National Agency for Quality Assessment and Accreditation (ANECA)

2024: Ramon y Cajal Fellowship: Tenure Track position by the Spanish Ministry of Science, Innovation and Universities (approx 250k€)

2023: Abilitazione scientifica nazionale II fascia: "Associate Professor" level recognize by the

Italian ministry of education (sector 02/A2)

2022: Professor Lector: “Assistant Professor” level recognized by the Agency for the Quality of the University system of Catalonia (AQU)

2021: Profesor Ayudante Doctor: “Assitant Professor” level recognized by the Spanish National Agency for Quality Assessment and Accreditation (ANECA)

2013: Highest Distinction: Bachelor’s Thesis: Numerical resolution of a system with coupled differential equations: applied to Quantum Scattering Problems with Internal Degrees of Freedom

2009: Third Prize: “IV Concurso Nacional para promocion de Jovenes Escritores Cientifico-Tecnicos”, ACTA-CEDRO, Scientific Writing

2008: First Prize: “I Concurso Narrativa Juvenil de la Comarca de La Vega”, Asociación Gran Vega de Sevilla, Creative Writing

Languages

Spanish: Native

English: Fluent

PhD studies carried out in Durham (United Kingdom)

Italian: Fluent

University level courses taught and students supervised in Italian

French: Basic knowledge

High school, currently living in Geneva

Japanese: Basic knowledge

A1.2 level certified

Conference Talks and Invited Seminars

DIS2026

Global analyses of PDFs

Bologna (Italy)

May 2026

Invited Seminar

The methodological landscape of PDF determinations

Michigan State University (USA, Virtual)

April 2026

SHARP First Network Conference

Machine Learning applications for PDFs: Recent Progress and Open Challenges

Curia (Portugal)

March 2026

NNPDF Collaboration meeting

Code, theory and data status towards NNPDF4.1

Amsterdam (The Netherlands)

February 2026

The world of possibilities of the NNPDF framework

The world of possibilities of the NNPDF framework

Milano (Italy)

December 2025

Invited Seminar

Challenges and developments in the determination of Parton Distribution Functions

Cambridge University (UK)

March 2025

Uncovering New Laws of Nature at the EIC

PDF determination and the EIC: impact and opportunities

Brookhaven National Lab (USA)

November 2024

QCD@LHC 2024

PDF constraints from new data and expectations from future experiments

Freiburg (Germany)

October 2024

NNPDF Collaboration Meeting

State of the Code

Morimondo (Italy)

September 2024

Invited Seminar <i>Challenges and developments on PDF determination</i>	IFIC Valencia (Spain) September 2024
DIS2024 <i>Phenomenological implications of modern PDF determinations</i>	Grenoble (France) April 2024
NNLOJET Collaboration meeting <i>NNLO Grids in NNPDF from NNLOJET</i>	Milano (Italy) March 2024
NNPDF Collaboration meeting <i>Code status and data implementation updates towards NNPDF4.1</i>	Amsterdam (The Netherlands) February 2024
Milan Christmas Meeting 2023 <i>Towards a framework for GPU event generation</i>	CERN, Switzerland December 2023
Collider Cross Talk <i>Why are we still talking about PDFs?</i>	CERN, Switzerland December 2023
PDF4LHC 2023 <i>Implications of NNPDF4.0 for LHC physics</i>	CERN, Switzerland November 2023
Event generator' and N(n)LO codes' acceleration <i>Towards a framework for GPU event generation</i>	CERN, Switzerland November 2023
NNPDF Collaboration Meeting <i>Status of the NNPDF framework and data implementation</i>	Gargnano, Lake Garda (Italy) September 2023
FPF Theory Workshop <i>Physics with Muons at the FPF (SM pow)</i>	CERN, Switzerland September 2023
LHCP11 2023 <i>Recent results on PDF extractions</i>	Belgrade, Serbia May 2023
HEP Theory Seminar <i>NNPDF4.0 and the path to reliable uncertainties</i>	Brookhaven National Lab. (USA, Virtual) May 2023
QCD@LHC 2022 <i>Theory developments in PDF determination</i>	IJCLab Orsay, France November 2022
QCD Seminar <i>NNPDF4.0 and the path to reliable uncertainties in PDF determination</i>	CERN, Switzerland November 2022
Invited seminar <i>GPU accelerated particle physics</i>	Nikhef, Amsterdam (The Netherlands) September 2022
NNPDF Collaboration Meeting <i>Status of the NNPDF fitting framework and theory pipeline</i>	Gargnano, Lake Garda (Italy) September 2022
Invited seminar <i>Facilitating GPU acceleration for Monte Carlo simulations</i>	Freiburg (Germany) July 2022
41th International Conference on High Energy Physics, ICHEP 2022 <i>MadFlow: automating Monte Carlo simulation on GPU for particle physics</i>	Bologna (Italy) July 2022
Transversity 2022 <i>Machine Learning in PDF determination: NNPDF4.0</i>	Pavia (Italy) May 2022
Invited seminar <i>Accelerating Monte Carlo simulations across hardware platforms</i>	USM/LMU Munich (Germany) May 2022

Invited seminar, Dalitz series <i>NNPDF4.0: the path to proton structure at 1% accuracy</i>	Oxford (UK, Virtual) November 2021
The 2021 International Workshop on the High Energy Circular Electron Positron Collider <i>GPU acceleration in High Energy Physics</i>	Nanjing (China, Virtual) November 2021
Invited Seminar (virtual) <i>Towards a GPU future for particle physics Monte Carlo simulations</i>	KIT Karlsruhe (Germany) June 2021
25th International Conference on Computing in High-Energy and Nuclear Physics (vCHEP) <i>MadFlow: towards the automation of Monte Carlo simulation on GPU for particle physics</i>	Virtual May 2021
PDF4LHC 2021 <i>New studies from the NNPDF group</i>	Virtual March 2021
Milano Joint Phenomenology Seminar <i>Offloading Monte Carlo simulations to hardware accelerators</i>	Milan (Italy, Virtual) February 2021
Invited Seminar (virtual) <i>PDF determination with a quantum hardware</i>	IFIC Valencia (Spain) February 2021
HSF WLCG Virtual Workshop <i>PDF/Vegas-Flow</i>	Virtual meeting November 2020
Generator Infrastructure and Tools Subgroup Meeting <i>VegasFlow and PDFFlow: accelerating Monte Carlo simulation across multiple devices (joint talk with M. Rossi)</i>	CERN (Virtual meeting) October 2020
40th International Conference on High Energy Physics, ICHEP <i>VegasFlow: accelerating Monte Carlo simulation across platforms</i>	Prague (Virtual meeting) August 2020
NNPDF Collaboration meeting <i>Optimizing the hyperoptimization</i>	Amsterdam (The Netherlands) February 2020
Artificial Intelligence for Science, Industry and Society Symposium (AISIS 2019) <i>Studying the parton content of the proton with deep learning models</i>	Ciudad de Mexico (Mexico) October 2019
James Stirling Memorial Conference & PDF4LHC <i>Methodological improvements in PDF determination</i>	Durham (UK) September 2019
NNPDF Collaboration meeting <i>n3fit and hyperoptimization in the context of NNPDF 4.0</i>	Varenna (Italy) August 2019
QCD@LHC 2019 <i>Towards a new generation of PDFs with deep learning models</i>	Buffalo, New York (USA) July 2019
NNLOJET Collaboration meeting <i>Numerical Integration with Neural Networks</i>	Zurich (Switzerland) May 2019

NNPDF Collaboration meeting <i>N3PDF studies of new methodologies</i>	Amsterdam (The Netherlands) February 2019
NNPDF Collaboration & N3PDF Kickoff Meeting <i>Recent developments within NNLOJET</i>	Gargnano, Lake Garda (Italy) September 2018
Loops and Legs in Quantum Field Theory 2018 <i>NNLO corrections to VBF Higgs boson production</i>	St. Goar (Germany) May 2018
HiggsTools Final Meeting <i>NNLO phenomenology with Antenna Subtraction</i>	Durham (UK) September 2017
Internal Seminar ϕ_η^* observable for Higgs production	Durham (UK) May 2017
Student Seminar <i>Higgs phenomenology with antenna subtraction</i>	Durham (UK) February 2017
Invited Seminar <i>Higgs phenomenology with antenna subtraction</i>	Valencia (Spain) January 2017
HiggsTools Second Annual Meeting <i>NNLO calculations for Higgs processes</i>	Granada (Spain) April 2016
Internal Seminar <i>Renormalisation Scale Dependence as a Testing Ground for NNLO calculations</i>	Durham (UK) February 2016
Student Seminar <i>Building and Playing with NNLO Monte Carlos</i>	Durham (UK) February 2016
HiggsTools First Annual Meeting <i>NNLO predictions for Higgs production at LHC</i>	Freiburg (Germany) April 2015

Student supervision

Co-director of bachelor Thesis <i>Correlation between statistical and physical properties of ensembles of Parton Distribution Functions, E. Stabilini</i>	University of Milan (Italy) 2021-2022
Co-director of master Thesis <i>Improving performance of automated generation of matrix elements for Monte Carlo event generators, G. Palazzo</i>	University of Milan (Italy) 2020-2021
Co-director of bachelor Thesis <i>Overfitting and gaussianity of Parton Distribution Functions, F.P. Guerci</i>	University of Milan (Italy) 2020-2021
Co-director of bachelor Thesis <i>The effect of discrete dataset on the gluon PDF, D. Chemoli</i>	University of Milan (Italy) 2020-2021
Co-director of master Thesis <i>New Monte Carlo Algorithms for Multi-Dimensional Integration with Hardware Acceleration, A. Pasquale</i>	University of Milan (Italy) 2020-2021
Co-director of master Thesis <i>Optimized regression models for parton distribution functions determination using deep learning models, N. Lambri</i>	University of Milan (Italy) 2019-2020

Co-director of master Thesis*Investigating GPU hardware for fast PDF convolutions, E. Villa***University of Milan (Italy)**

2018-2019

Co-director of bachelor Thesis*Stability in the determination of parton distributions, F. Settimo***University of Milan (Italy)**

2018-2019

Publications

- [1] Juan M. Cruz-Martinez, Tommaso Giani, and Lucian A. Harland-Lang. “Assessing the impact of fitting methodology at aN^3LO with FPDF: an open source tool for extracting Parton distribution functions in the Hessian approach”. In: *JHEP* 07 (2026), p. 185. DOI: 10.1007/JHEP07(2026)185. arXiv: 2602.07118 [hep-ph].
- [2] Richard D. Ball et al. “A determination of $\alpha_s(m_Z)$ at $\text{aN}^3\text{LO}_{\text{QCD}} \otimes \text{NLO}_{\text{QED}}$ accuracy from a global PDF analysis”. In: *Eur. Phys. J. C* 85.9 (2025), p. 1001. DOI: 10.1140/epjc/s10052-025-14676-y. arXiv: 2506.13871 [hep-ph].
- [3] Alexander Huss et al. “NNLOJET: A parton-level event generator for jet cross sections at NNLO QCD accuracy”. In: *SciPost Phys. Codeb.* 69 (2026), p. 1. DOI: 10.21468/SciPostPhysCodeb.69. arXiv: 2503.22804 [hep-ph].
- [4] Thomas Cridge et al. “Combination of aN^3LO PDFs and implications for Higgs production cross-sections at the LHC”. In: *J. Phys. G* 52 (2025), p. 6. DOI: 10.1088/1361-6471/adde78. arXiv: 2411.05373 [hep-ph].
- [5] Amedeo Chiefa et al. “Parton distributions confront LHC Run II data: a quantitative appraisal”. In: *JHEP* 07 (2025), p. 067. DOI: 10.1007/JHEP07(2025)067. arXiv: 2501.10359 [hep-ph].
- [6] Juan Cruz-Martinez et al. “NNPDFpol2.0: a global determination of polarised PDFs and their uncertainties at next-to-next-to-leading order”. In: *JHEP* 07 (2025), p. 168. DOI: 10.1007/JHEP07(2025)168. arXiv: 2503.11814 [hep-ph].
- [7] Juan M. Cruz-Martinez, Giuseppe De Laurentis, and Mathieu Pellen. “Accelerating Berends–Giele recursion for gluons in arbitrary dimensions over finite fields”. In: *Eur. Phys. J. C* 85.5 (2025), p. 590. DOI: 10.1140/epjc/s10052-025-14318-3. arXiv: 2502.07060 [hep-ph].
- [8] Juan Cruz-Martinez, Alexander Huss, and Christopher Schwan. “Fast interpolation grids for the Drell–Yan process”. In: *Eur. Phys. J. C* 85.4 (2025), p. 459. DOI: 10.1140/epjc/s10052-025-14170-5. arXiv: 2501.13167 [hep-ph].
- [9] Juan Cruz-Martinez et al. “Hyperparameter optimisation in deep learning from ensemble methods: applications to proton structure”. In: *Mach. Learn. Sci. Tech.* 6.2 (2025), p. 025027. DOI: 10.1088/2632-2153/adcd39. arXiv: 2410.16248 [hep-ph].
- [10] Matteo Robbiati, Juan M. Cruz-Martinez, and Stefano Carrazza. “Determining probability density functions with adiabatic quantum computing”. In: *Quantum Machine Intelligence* 7.1 (2025), p. 5. DOI: 10.1007/s42484-024-00228-2. arXiv: 2303.11346 [quant-ph].
- [11] Juan Cruz-Martinez et al. “LO, NLO, and NNLO parton distributions for LHC event generators”. In: *JHEP* 09 (2024), p. 088. DOI: 10.1007/JHEP09(2024)088. arXiv: 2406.12961 [hep-ph].
- [12] Richard D. Ball et al. “The path to N^3LO parton distributions”. In: *Eur. Phys. J. C* 84.7 (2024), p. 659. DOI: 10.1140/epjc/s10052-024-12891-7. arXiv: 2402.18635 [hep-ph].

- [13] Richard D. Ball et al. “Determination of the theory uncertainties from missing higher orders on NNLO parton distributions with percent accuracy”. In: *Eur. Phys. J. C* 84.5 (2024), p. 517. DOI: 10.1140/epjc/s10052-024-12772-z. arXiv: 2401.10319 [hep-ph].
- [14] Richard D. Ball et al. “Photons in the proton: implications for the LHC”. In: *Eur. Phys. J. C* 84.5 (2024), p. 540. DOI: 10.1140/epjc/s10052-024-12731-8. arXiv: 2401.08749 [hep-ph].
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- [17] Juan M. Cruz-Martinez et al. “The LHC as a Neutrino-Ion Collider”. In: *Eur. Phys. J. C* 84.4 (2024), p. 369. DOI: 10.1140/epjc/s10052-024-12665-1. arXiv: 2309.09581 [hep-ph].
- [18] Andrea Barontini et al. “Pineline: Industrialization of high-energy theory predictions”. In: *Comput. Phys. Commun.* 297 (2024), p. 109061. DOI: 10.1016/j.cpc.2023.109061. arXiv: 2302.12124 [hep-ph].
- [19] Richard D. Ball et al. “Evidence for intrinsic charm quarks in the proton”. In: *Nature* 608.7923 (2022), pp. 483–487. DOI: 10.1038/s41586-022-04998-2. arXiv: 2208.08372 [hep-ph].
- [20] Stefano Carrazza, Juan M. Cruz-Martinez, and Roy Stegeman. “A data-based parametrization of parton distribution functions”. In: *Eur. Phys. J. C* 82.2 (2022), p. 163. DOI: 10.1140/epjc/s10052-022-10136-z. arXiv: 2111.02954 [hep-ph].
- [21] A. Buckley et al. “A comparative study of Higgs boson production from vector-boson fusion”. In: *JHEP* 11 (2021), p. 108. DOI: 10.1007/JHEP11(2021)108. arXiv: 2105.11399 [hep-ph].
- [22] Richard D. Ball et al. “The path to proton structure at 1% accuracy: NNPDF Collaboration”. In: *Eur. Phys. J. C* 82.5 (2022), p. 428. DOI: 10.1140/epjc/s10052-022-10328-7. arXiv: 2109.02653 [hep-ph].
- [23] NNPDF Collaboration. “An open-source machine learning framework for global analyses of parton distributions”. In: *Eur. Phys. J. C* 81.10 (2021), p. 958. DOI: 10.1140/epjc/s10052-021-09747-9. arXiv: 2109.02671 [hep-ph].
- [24] Stefano Carrazza et al. “MadFlow: automating Monte Carlo simulation on GPU for particle physics processes”. In: *Eur. Phys. J. C* 81.7 (2021), p. 656. DOI: 10.1140/epjc/s10052-021-09443-8. arXiv: 2106.10279 [physics.comp-ph].
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